Orchestrating Microbiome Analysis with Bioconductor

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Abstract

The expansion of microbiome research has led to the accumulation of interlinked datasets encompassing versatile taxonomic and functional assays. The analysis of increasingly large and heterogeneous multi-modal microbiome data would benefit from unified approaches supporting the design of modular data science workflows through interoperable methods. The Bioconductor project recently developed an optimized statistical programming framework for multi-assay data integration. Building on this foundation, we introduce a community-developed, open-source framework for microbiome data science. In contrast to the previous alternatives, the methodology is specifically designed to support joint analysis of hierarchical, interlinked, and heterogeneous multi-table datasets that are increasingly common in modern microbiome research. This framework encompasses open data, methods, tutorials, and an active online community, thereby supporting standardized and reproducible data wrangling, joint analysis, and reporting. We have detailed the functionality and usage in the OMA book (https://bioconductor.org/books/release/OMA), which offers guidance for prospective users and contributors.

Keywords: microbiome, Bioconductor, data science, data integration

Introduction

Modern data science applications critically rely on open-source methods created by the research community[1;2]. Open software ecosystems have become central innovation hubs, driving technological and cultural transformation[3]. The Bioconductor project has become a significant distribution channel for open data science methods in the life sciences. Its vast developer community maintains over 2,300 quality-controlled R packages for various fields of life science informatics[4]. Additionally, Bioconductor has evolved into a global community that supports data science collaboration and training[5], playing an increasingly important role in advancing computational skills that are essential for modern microbiology education[6].

Microbiome research focuses on how microbial communities vary, function and interact with their environment and a potential host[7]. The technologies used in this field generate vast amounts of high-dimensional and hierarchically structured data, commonly in the form of microbial DNA sequences detected in a set of samples. Following the rapid expansion of microbiome research and technological advances, metagenomic sequencing is complemented by other types of high-throughput measurements. This has led to the accumulation of multi-modal datasets—collections that combine different types of molecular data. These interlinked datasets encompass diverse taxonomic and functional assays, ranging from microbial abundances and phylogenetic relationships to functional profiles. These are summarized into tabular data sets whose typical sizes range up to thousands of selected features and samples for downstream statistical analyses.

Microbiome data scientists are thus increasingly facing the practical challenges of integrating heterogeneous and high-dimensional obsevations across multiple data modalities into scalable and reproducible workflows[8–10]. The unique statistical properties and complexity of microbiome data necessitate specialized approaches[7;11–14], yet the diversity of proposed solutions and inconsistent outcomes[15–17] can make it difficult to identify the optimal methods and build interoperable workflows[18;19]. Accordingly, there is a need for standardization[7].

Open microbiome data science *frameworks* have emerged to provide sets of interoperable methodologies in specific computational environments[20–23], and a broader ecosystem of digital resources, engaged user communities and collaboration culture that extend the capacities far beyond the underlying technical framework. However, frameworks focused solely on microbiome data science fall short on addressing the need to integrate data across multiple modalities, such as metagenomics, metabolomics, and host genomics.

Bioconductor integrates the methods of microbiome data science into a wider statistical programming framework, where interoperable methods for single-cell omics[24], spatial transcriptomics[25], metabolomics and proteomics[26], and other fields are concurrently developed.

This approach can introduce some challenges, such as dependency management, package compatibility, and differences in package stability. These issues are addressed through coordinated mechanisms that help to ensure the robustness of the software ecosystem. First, submitted packages undergo peer review, ensuring relevance and adherence to standards such as unit tests, reproducible documentation, and working examples. Secondly, Bioconductor follows a synchronized, semiannual release cycle, during which packages are frozen and mutually compatible. Thirdly, all packages are continuously tested within Bioconductor's automated build system, which ensures that breaking changes are rapidly detected before release. Each package has an active maintainer who is notified of failures and responsible for resolving them in a timely manner. Finally, community guidelines and continuous integration practices, together with shared data structures such as (Tree)SummarizedExperiment, further improve interoperability between packages.

Here, we introduce a stable and optimized data science ecosystem supporting the integration and analysis of multi-modal datasets in microbiome research. It provides harmonized data import and representation, and fast, robust methods for transforming abundance tables into biological insights. These methods have been extensively tested and distributed through R/Bioconductor packages (https://bioconductor.org/packages), supporting the community-driven development of the framework and integration within the broader Bioconductor ecosystem. To promote evidence-based best practices, we introduce the online book *Orchestrating Microbiome Analysis with Bioconductor* (OMA), which represents a comprehensive compilation of established approaches and emerging trends in microbiome data science using the R/Bioconductor ecosystem.

Data infrastructure

Data scientists routinely deal with interlinked assays, hierarchical data (*e.g.*, ontologies, trees), relational data (*e.g.*, networks), and supporting side information. This side information can include sample attributes such as treatment group, collection site, or time point. The ability to integrate heterogeneous data elements is essential in biological research, where the individual components of a system can seldom be studied in isolation. However, this growing complexity introduces additional overhead, as researchers must track samples and features across multiple tables and manage an increasing number of data elements and non-trivial links between them. Moreover, input data formats vary across methods, leading to technical complications and time lost on data wrangling. Bioconductor is addressing these challenges through standardized data structures that are supported by an ecosystem of quality-controlled, interoperable methods. Adopting the framework will thus allow the analyst to invest more time in core analytical tasks.

Bioconductor data structures support statistical analysis of complex data combinations[27] and are widely adopted in omics research (Figure 1). They have been implemented through specialized object-oriented classes that integrate multiple data elements into a single structure known as a *data container*. The SummarizedExperiment (SE) class[27] is the primary data container underlying the Bioconductor ecosystem (Figure 2 a and Table 1). Its position among the top 1% most-downloaded packages in Bioconductor reflects its widespread recognition and increasing adoption among developers. SE provides a standardized solution to store tabular data by linking numeric abundance matrices with side information on features or rows (*e.g.*, taxonomy), and samples or columns (*e.g.*, origin, collection time, host health).

The interoperability of SE with other Bioconductor data structures enables the transfer of methods across different fields of omics research, thereby promoting the development of modular and scalable data science methods and ultimately yielding improved overall quality control and user support[5]. In particular, the SE framework has been extended to similar data integration tasks, notably in single-cell and spatial omics through the SingleCellExperiment (SCE)[24] and SpatialExperiment[25;28] classes.

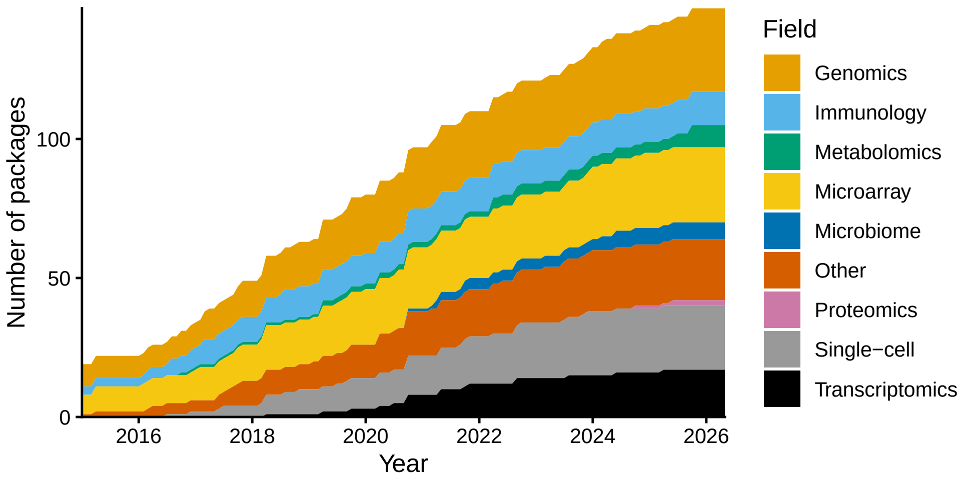

Figure 1: **Adoption of the SummarizedExperiment (SE) data container among Bioconductor developers.** The growth in the number of Bioconductor packages supporting SE over time is shown, categorized by their respective fields. Packages are counted by the date they were added to Bioconductor. Some packages added the SE support after their original release.

Hierarchical data structures. Microbiome data scientists frequently need to integrate information on feature phylogenies and sample hierarchies to accurately reflect fine-scale variability in microbiome data. While maintaining interoperability with the broader Bioconductor ecosystem, the TreeSummarizedExperiment (TreeSE) class[29] extends the SE and SCE data structures to hierarchical datasets by incorporating both feature and sample organizations as tree structures (Figure 2 b and Table 1). Beyond the core components of microbiome datasets, including taxa abundances, sample metadata, phylogenetic trees, and reference sequences, TreeSE provides a flexible foundation for multi-modal data integration, encompassing functional predictions, resistome, metabolome, and other molecular profiles, as well as alternative feature representations. It also stores results from common analytical procedures, such as dimensionality reduction, statistical transformations, and agglomeration by taxonomic, functional, or other information. By inheriting full compatibility with SE and SCE, TreeSE enables scalable analyses and the development of modular, interoperable workflows across the Bioconductor ecosystem.

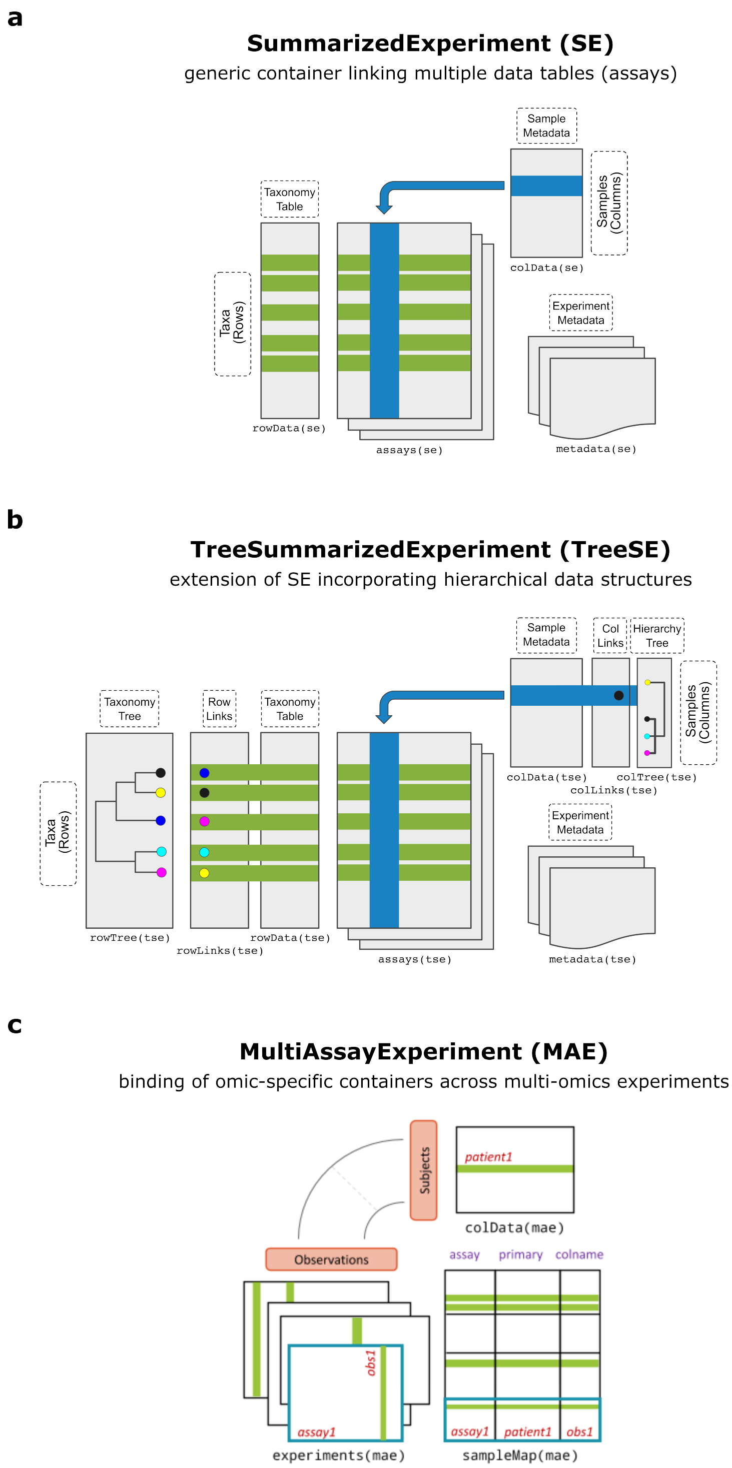
a
SummarizedExperiment (SE)
generic container linking multiple data tables (assays)
Sample Metadata
Samples (Columns)
Taxonomy Table
colData(se)
Taxa (Rows)
Experiment Metadata
rowData(se)
assays(se)
metadata(se)
b
TreeSummarizedExperiment (TreeSE)
extension of SE incorporating hierarchical data structures
Sample Metadata
Col Links
Hierarchy Tree
Taxonomy Tree
Row Links
Taxonomy Table
Samples (Columns)
colData(tse)
colTree(tse)
colLinks(tse)
Taxa (Rows)
Experiment Metadata
rowTree(tse)
rowData(tse)
rowLinks(tse)
assays(tse)
metadata(tse)
c
MultiAssayExperiment (MAE)
binding of omic-specific containers across multi-omics experiments
Subjects
patient1
colData(mae)
Observations
assay
primary
colname
obs1
assay1
assay1
patient1
obs1
experiments(mae)
sampleMap(mae)

Figure 2: **Integrative Bioconductor containers for microbiome data. (a) SummarizedExperiment (SE)**[27] is a generic container for single-omics experiments. It accommodates *assays* that describe the feature abundances in each sample (*e.g.*, taxonomic units, resistance genes, or predicted functions), and tabular side information on each sample (*colData*) and feature (*rowData*), *e.g.*, the taxonomic classification. **(b) TreeSummarizedExperiment (TreeSE)**[29] supports hierarchical, multi-table microbiome analysis. In addition to the SE elements, TreeSE incorporates tree information on feature hierarchies, such as phylogenetic relations between the taxonomic units (*rowTree*) or host species (*colTree*), and results of dimensionality reduction (*reducedDims*). TreeSE can thus link multiple abundance tables and their transformed versions (*e.g.*, statistical transformation, taxonomic agglomeration, or dimensionality reduction) with tabular and hierarchical side information on the features (rows) and samples (columns). Additional slots for reference sequences and experiment metadata are available. **(c) MultiAssayExperiment (MAE)**[30] facilitates the integration of interlinked datasets representing different omics modalities linked by potentially non-trivial sample mappings. Adapted from[30].

Multi-assay data structures. Contemporary microbiome research frequently involves data across multiple measurement modalities for enhanced functional and mechanistic insights. This brings the need to find an appropriate representation for each data type and map the links between them. Whereas the TreeSE data container is broadly applicable to different modalities such as taxonomic, resistome, transcriptome, proteome, and metabolome profiling data, dedicated data containers have been developed for certain modalities by the Bioconductor community to address their specific characteristics; examples include single-cell sequencing[24] and host genomics[31]. Linking data across modalities presents an additional technical challenge, as multi-omics datasets are often sparse and only a subset of the samples may be shared across multiple modalities[32]. In other cases, a single sample in one modality may link to several samples in another modality[33] (*e.g.*, due to technical or spatial replication). These scenarios require the ability to integrate multiple data modalities into a single object and create flexible sample mappings across them. The MultiAssayExperiment (MAE) data container[30] addresses these challenges by introducing a formal mapping layer (sampleMap) that links biological samples to their corresponding measurements in each modality (Figure 2 c and Table 1). This design supports one-to-one, one-to-many, and partially overlapping sample relationships across experiments. Importantly, this mapping is also accessible for downstream operations. For instance, when subsetting a MAE object (*e.g.*, selecting samples with both microbiome and metabolome data), the framework automatically resolves the intersection of available samples and ensures consistent alignment across all modalities. Such structured handling of sample relationships is particularly important for integrative methods, as it reduces the risk of sample mismatches, a common source of irreproducibility in ad hoc multi-table analyses. For instance, diagonal integration methods[34]—where shared features across data types may not be explicitly defined—can benefit from such a systematic framework that enforces anchor alignment. This reflects lessons learned from single-cell data integration, where shared anchors and hierarchical mapping have proven essential for interpretability. To summarize, flexible mapping of samples across several data objects is key for efficient data integration, method sharing, and overall interoperability.

Table 1: Key elements of the Bioconductor data containers for microbiome analysis.

Slot	Content	SE	TreeSE	MAE[a]
Tables				
assays	tables of features (rows) and samples (cols)	X	X	X
altExps[b]	experiments with variable number of features		X	X
reducedDims	results of dimensionality reduction		X	X
Rows				
rowData	side information on features	X	X	X
rowPairs	linkage between pairs of associated features		X	X
rowTrees	hierarchical organization of features		X	X
referenceSeq	reference sequences of features		X	X
Columns				
colData	side information on samples	X	X	X
colTrees	hierarchical organization of samples		X	X
sampleMap[c]	flexible sample mapping across experiments			X
Other				
metadata	additional information on the experiment	X	X	X

[a]While MAE does not contain conventional slots (with the exception of colData), its experiments can include any number of data containers that provide those slots. [b]altExp samples must match assay samples in a 1-to-1 mapping. [c]The sampleMap supports diverse mappings between samples from different experiments (1-to-1, 1-to-many, many-to-1 and many-to-many). SE, SummarizedExperiment; TreeSE, TreeSummarizedExperiment; MAE, MultiAssayExperiment.

Data resources. Several open microbiome data resources are supported, enabling direct data import into the aforementioned data containers for further downstream analysis within the Bioconductor methods ecosystem (Table 2). Examples of the available data resources include curatedMetagenomicData[35], HoloFood[36], microbiomeDataSets (Bioconductor package), and the EBI/MGnify database[37]. In addition, Bioconductor packages often include built-in demonstration datasets. Collectively, these resources encompass taxonomic and functional profiles from thousands of published microbiome studies and tens of thousands samples that can be accessed with the available tools for research and educational purposes.

Table 2: Open microbiome data resources (last accessed on 05/07/2026).

Data resource / Package	References	# Studies	# Samples
MGnifyR[a]	[37]	5,616	429,108
Metalog[b]	[38]	1,074	157,483

Data resource / Package	References	# Studies	# Samples
curatedMetagenomicData[a]	35	93	22,588
miaData[c]		9	20,404
microbiomeDataSets[a]		6	19,100
HMP2data[a]		3	11,493
HoloFoodR[a,d]	36;39	9	9,990
MicrobiomeBenchmarkData[a]	17	6	1,125

[a]Bioconductor package with the corresponding name. [b]Data retrieval to TreeSE format instructed in OMA online book. [c]Zenodo mia community. [d]HoloFood microbiome datasets are deposited in the MGnify database. HMP, Human Microbiome Project.

Alternative frameworks

Several alternative frameworks to microbiome analysis are available.
Within Bioconductor, phyloseq[23] provides widely adopted tools with a primary focus on 16S amplicon data analysis. (Tree)SE accommodates all core components of phyloseq objects, including abundance tables, taxonomic annotations, phylogenetic trees, reference sequences, and sample metadata. The advantages of newer (Tree)SE container include many additional data elements and support for multi-assay analyses, enhanced scalability, and a better integration with the overall Bioconductor ecosystem (see Section 23).
Widely used frameworks beyond R/Bioconductor include several command-line toolkits. Mothur[21] is primarily designed for 16S amplicon analysis and less well suited for generic microbiome and integrative multi-omic analyses. QIIME 2[22] is a Python-based, AI-ready microbiome multi-omics data science platform. It primarily targets amplicon and metagenomic analyses, and incorporates methods from scikit-bio Python library[40] which supports multi-omic integration tasks. The overall interoperability with independently developed omics tools may remain more limited, however. A particular and timely advantage of these frameworks is their direct connection with the strong machine learning and AI ecosystem in Python. Anvi'o[20] is a complementary, community-driven analysis and visualization platform for microbial 'omics with a distinct focus on genome-resolved metagenomics and detailed sequence-level analyses. Whereas each of these tools is supported by active user and developer communities, their overall design is different. In contrast, Bioconductor emphasizes broad interoperability and shared statistical programming infrastructure with other areas of omics research, primarily serving the R community.
Benchmarking the scalability of routine operations applied to subsets of the Metalog database[38] shows substantial performance gains for the (Tree)SE framework over alternatives (Figure 3 and Supplementary Figure 1). Overall, (Tree)SE tends to scale most efficiently in terms of execution time and allocated memory as feature and sample sizes increase. It is applicable to abundance matrices of up to 100,000 samples

and 10,000 features on a standard laptop, except the heaviest operations such as phylogenetic beta diversity (Unifrac) calculation, where the general break point is reached around 10,000 samples with all methods. At a feature set size to 10,000 the break point is additionally reached for phILR transformation and melting operation. At smaller sample sizes phyloseq or its accelerated extension speedyseq are occasionally faster but these differences are practically negligible (few seconds on a standard laptop). QIIME 2 is generally slower but the differences become smaller with increasing sample sizes. The enhanced scalability in (Tree)SE benefits, for instance, from the built-in support for sparse and delayed matrices in SE-based data containers. Despite these advantages, further optimizing time and memory consumption represents a key area for further development as discussed in Section 4.
To facilitate migration from alternative frameworks, the OMA book provides conversion tables and utility functions to convert between TreeSE and many external data formats.

Data preparation

While several tools for microbiome data analysis are available[11;12;41], their implementations typically differ in terms of expected input and output formats. Common data standards promote modular data science workflows, speeding up methods development, benchmarking, and replication. Building upon these principles, we describe a set of interoperable software libraries that support microbiome data integration and analysis based on the two integrative data containers, TreeSE and MAE (Figure 4).
This framework specifically supports the downstream analysis of high-throughput sequencing data from microbiome experiments after summarizing the original measurements into abundance tables (*e.g.*, taxonomic or functional profiles). Taxonomic abundance tables are one of the most commonly encountered data types. Common measurement platforms include amplicon-based marker gene studies (*e.g.*, 16S rRNA) and metagenomic sequencing, but alternative profiling techniques are available including phylogenetic microarrays[42] and high-density optical mapping[43]. The interoperability of the data containers support modular workflow design for integrating different omics data types into the broader Bioconductor ecosystem. This mitigates reproducibility challenges posed by rapidly evolving reference databases and profiling tools (*e.g.*, MetaPhlAn) by enabling structured coexistence and direct comparison within a unified, coherent framework[44;45].
Data import. While the TreeSE data container can be populated with microbiome data from common tabular data types, non-standard file formats may require additional data wrangling. To streamline data import, we provide importers for a variety of standard microbiome file formats, including BIOM[46], MetaPhlAn/HUMAnN (versions 2-4)[47], Mothur[21], QIIME 2[22], and the TAXonomic Profile Aggregation and standardization format (taxpasta)[48]. External to Bioconductor, the qiime2R package provides additional functionality to create TreeSE objects directly from QIIME 2 artifacts. These importers bridge the gap between the original data processing, other tools, and downstream statistical analysis in R, allowing researchers to focus on interpreting biological patterns rather than wrangling file formats. Moreover, converters between alternative data structures in R are available supporting seamless conversions between

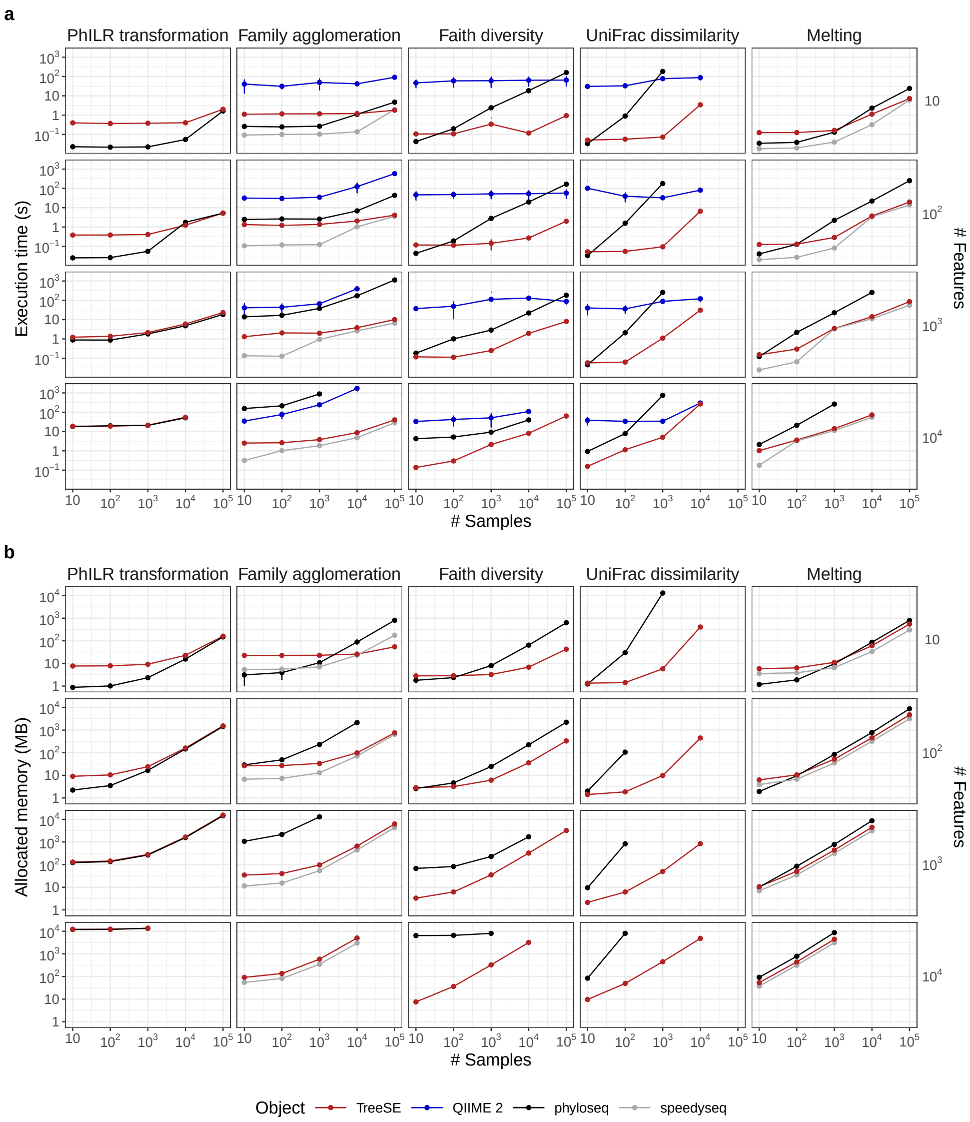

Figure 3: **Execution time and allocated memory for routine microbiome data operations.** TreeSummarizedExperiment, QIIME 2 and phyloseq—with its extension speedyseq—are three popular frameworks for microbiome analysis. We benchmarked them for five routine operations applied to random subsets of features (rows) and samples (columns) from the Metalog database[38]. Both execution time (s) and allocated memory (MB) were measured as the mean value over ten replicates with varying subsets of the same dimension. Errorbars show the 95% confidence interval of the performance estimate. The benchmarking was conducted in R using the bench library with 4 CPU cores and 16 GB RAM. QIIME 2 lacks allocated memory estimates because memory cannot be monitored for external processes. The source code for the benchmarking analysis is openly available in the Zenodo mia community.

TreeSE and BIOM, DADA2[9], and phyloseq (the mia::convert* functions). Similarly, microbiome data stored in TreeSE objects can also be exported to external services such as MicrobiomeDB[49] and other languages (*e.g.*, Julia). These converters facilitate widespread access to microbiome analysis methods. Imported datasets from different omics modalities can then be interlinked within the MAE data container with its flexible sample mappings as described above. This integrated data object can then be analyzed with dedicated downstream methods.

Quality control and exploration. After importing the data into R, the recommended first step involves basic data exploration and quality control[50]. In addition to methods for identifying contaminant sequences and calculating summary statistics, the framework includes a versatile set of custom visualization methods to assess data variability, outliers, homogeneity of variance and other aspects that may need to be considered in downstream data analysis. The mia and miaViz packages provide convenient access to many operations for the integrative data containers regarding common tasks in microbiome data exploration and visualization (Figure 5).

Filtering and agglomeration. Microbiome datasets can contain tens of thousands of unique features depending on the sequencing resolution, used reference databases, and the scope of analysis. Moreover, the integration of multiple omics modalities adds another level of challenges. To summarize data into biologically meaningful groups or to reduce multiple comparisons the data is often subsetted, filtered or agglomerated to higher-level taxonomic or functional categories or stratified by sample groups. The package ecosystem, and the mia package in particular, feature various tools for such tasks. For instance, the prevalence-based methods can be helpful to remove rare features that are likely sequencing artifacts or present otherwise limited value for the analysis. The mechanism of alternative experiments (altExp) allows agglomeration by taxonomic levels or other features such as pathogenicity or functional potential (*e.g.*, butyrate production[51], gut-brain modules[52], and host-associated signatures as in the BugSigDB[53]). The agglomerated data can be incorporated as a new alternative experiment within the original data object. The automated tracking of feature and sample links ensures that changes are propagated across all relevant elements of the data object, such as transformed abundance tables, phylogenetic trees, or interlinked omics modalities. This helps to avoid redundant processing and enables efficient management of interlinked multi-modal datasets.

Transformation. Statistical transformations commonly used in microbiome research include relative abundance[54], tree-aware phylogenetic ILR (phILR)[55], centered log-ratio (CLR), and robust CLR (rCLR)[56]. The latter three aim to mitigate compositionality bias. A related concept that controls for unequal sampling depth is data rarefaction. Rarefaction is the process of repeatedly subsampling data to equal read depth and averaging computed metrics[57]. Subsampling data to equal read depth (*rarefying*) without iteration is not recommended[58–60]. Our framework supports these and a variety of other transformations commonly applied to microbiome data.

Data enrichment. Incorporating additional data, such as transformed feature abundance tables, is a frequent operation in applied data analysis, and it often takes place iteratively. The integrative data containers support gradual and organized accumulation of interlinked data elements including transformations, aggregated data, reduced dimensions, and derived data on samples and features. We generally recommend initializing the data objects at the beginning of the workflow, where they can then be gradually enriched as the project advances. Data can be enriched by importing new external information as well as by deriving new information from the data itself into the same data container as described in the next section. By systematically extending the upstream data preparation step—instead of ad-hoc calculations scattered amidst the workflow—one can achieve a well-organized and transparent provenance of the integrated data object.

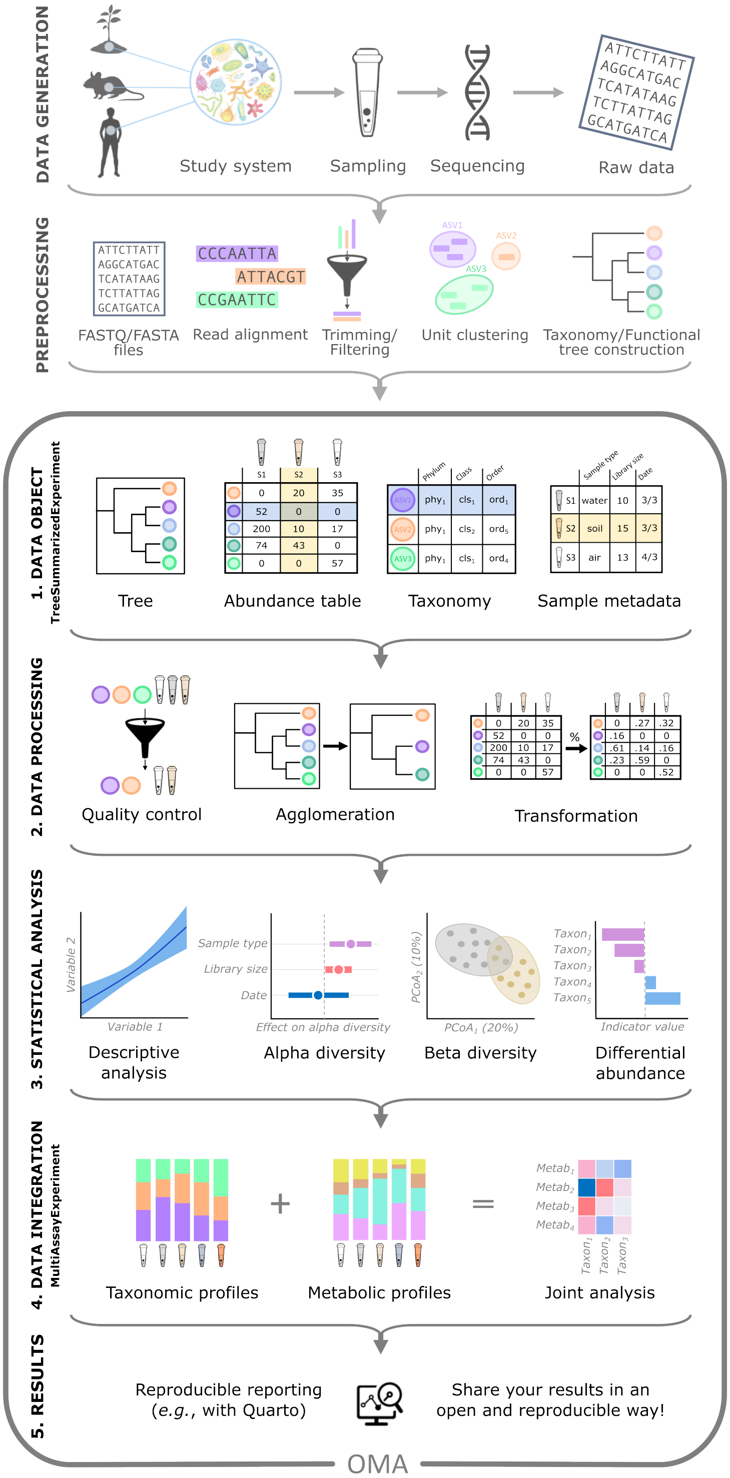
DATA GENERATION
Study system
Sampling
Sequencing
Raw data
PREPROCESSING
FASTQ/FASTA files
Read alignment
Trimming/ Filtering
Unit clustering
Taxonomy/Functional tree construction
1. DATA OBJECT
TreeSummarizedExperiment
Tree
Abundance table
Taxonomy
Sample metadata
2. DATA PROCESSING
Quality control
Agglomeration
Transformation
3. STATISTICAL ANALYSIS
Descriptive analysis
Alpha diversity
Beta diversity
Differential abundance
4. DATA INTEGRATION
MultiAssayExperiment
Taxonomic profiles
Metabolic profiles
Joint analysis
5. RESULTS
Reproducible reporting (e.g., with Quarto)
Share your results in an open and reproducible way!
OMA

Figure 4: **Multi-modal data integration workflow.** Following data generation and preprocessing (top two panels), the integrative framework covers five steps of microbiome data integration and analysis: 1) Data is combined into a TreeSummarizedExperiment (TreeSE) data container, storing information on the samples (tube icons) and their features (*e.g.*, taxonomic, functional, or resistome profiles; colored circles). TreeSE objects can accommodate multiple assay tables as well as additional hierarchical structures. 2) The TreeSE data object can then be processed to clean the data. This involves filtering out low-quality samples and features, transforming assays for example by normalizing feature-wise to enhance the comparability of sample-specific abundances, or agglomerating features to specific taxonomic levels or functional categories. 3) The cleaned data can then be analyzed using statistical models to uncover relations among samples (*e.g.*, with ordination methods), analyze drivers of variation in community diversity and composition, or test for differential abundance (*e.g.* between sampling time or location). 4) The framework supports the integration of various types of omics modalities through the MultiAssayExperiment and omic-specific data containers provided by Bioconductor. For example, the taxonomic and metabolomic profiles of samples can be summarized and compared in a joint analysis among features. 5) Finally, Quarto notebooks can aid reproducible analysis and reporting.

Downstream data analysis

This framework is designed to improve the management of integrative analysis workflows. It facilitates joint operations on interlinked data elements, helping to shift the focus towards multi-modal analyses and methods development while also supporting standard data analyses in microbiome research.

Microbiome data analysis often relies on specialized statistical approaches, since data sparsity, zero-inflation, compositionality, and heteroscedasticity pose challenges for standard methods[11;54]. Furthermore, microbiome data is hierarchically structured across taxonomic, ecological, temporal, and spatial levels[7;61]. Alternative processing pipelines introduce different biases and generate varying data formats; for example, the same sample can yield differing taxonomic profiles depending on the computational method used[62]. These and other unique characteristics have wide-reaching implications for microbiome analysis, necessitating dedicated analysis methods[7]. The shared data containers support community-driven development and reproducible benchmarking of methods.

Enhancing scalability is essential with increasingly large data sets. The computational efficiency can be improved by parallelization, out-of-memory analyses leveraging the built-in support for sparse and delayed matrices in SE-based data containers, more efficient implementations through code optimization, executing costly computations in C++, and methodological advancements, as in *e.g.*[63]. In many cases, such support can be readily inherited from the underlying Bioconductor infrastructure, rather than newly implemented in mia or other packages. For instance, the (Tree)SE and MAE containers are agnostic to assay representation and natively support sparse (dgCMatrix, SparseArray) and file-backed (HDF5Array, DelayedArray) assays provided by the Bioconductor ecosystem (see *e.g.*[27]). Many of the Bioconductor methods readily support parallelization, and missing capabilities can be inherited to dependent operations through argument passing or newly implemented. BiocParallel can support operations that decompose into independent blocks by samples, features, or other data elements. Non-decomposable operations, such as distance matrices and ordinations, must be realized in memory or parallelized by a dedicated algorithm (as in *e.g.* striped Unifrac[64]).

The following sections provide an overview of some of the currently available tools, encompassing standard methods for taxonomic and other abundance tables as well as integrative analyses between modalities.

Community indicators. Various general indicators are available to quantify the microbial community state or other sample properties. Community diversity represents a common measure in microbial ecology[13] (Figure 5). Alpha diversity metrics generally quantify the observed or estimated number of distinct taxa (*richness*) and their distribution (*evenness*). Many diversity indices (*e.g.*, Shannon index), combine these two aspects. Phylogenetic diversity metrics additionally account the breadth of phylogenetic relatedness among community members, but are more computationally demanding than abundance-based metrics. However, a new implementation, the Stacked Faith's Phylogenetic Diversity improves computational efficiency by 1–2 orders of magnitude[63] by an optimized single-pass algorithm and utilizing data sparsity. The framework supports analyses with rarefaction through the mia package[57]. Other global indices of community state include measures of dominance, rarity, skewness, kurtosis, modality, library size (number of reads), or absolute abundance quantification. Sample information in the TreeSE object (colData) can be enriched with these derived quantities. By default, the mia::addAlpha method returns a set of complementary alpha diversity measures. Alpha diversity indices have been frequently applied to measure not only taxonomic diversity but also resistome diversity[65], and potentially other modalities.

Community similarity is commonly evaluated with beta diversity measures, which quantify multivariate dissimilarity between feature communities across all pairs of samples and can be used to compare community composition quantitatively. The framework supports common dissimilarity metrics in microbial ecology, including Bray-Curtis, Jaccard, Aitchison[13], and the phylogeny-aware UniFrac[66]. Community dissimilarity is commonly visualized with unsupervised ordination methods that project the data onto a lower-dimensional space. Among unsupervised techniques, the most commonly used ones include Principal Component Analysis (PCA) and Principal Coordinate Analysis (PCoA), which encompasses both metric and non-metric multidimensional scaling (Figure 5). Statistical tests (*e.g.*, PERMANOVA) and supervised ordination methods, such as distance-based Redundancy Analysis (dbRDA), can complement the unsupervised analyses, and help uncover and quantify covariation between community composition and sample-specific covariates. This type of analysis can benefit from compositionally robust measures (*e.g.*, robust Aitchison distance[56]), or phylogeny-aware dissimilarity measures specifically developed for microbiome research (*e.g.*, UniFrac[66]). Sample dissimilarity provides the basis for many specialized microbiome techniques, including microbial community typing (clustering) with Dirichlet multinomial mixtures[67] and community state types[68], or factorization into community signatures[69].

These and other multivariate methods are provided by mia, miaViz, and additional single-cell packages (scater, scuttle[24]). Quantification of sample dissimilarities is a common task, and many omics modalities can benefit from shared, standardized methods provided by the integrative (Tree)SE framework.

Association analyses. Differential Abundance Analysis (DAA) and Differential Prevalence Analysis (DPA) methods identify indicator microbial features whose abundances (DAA) or prevalences (DPA) vary across study groups or continuous gradients, for example in terms of spatial or temporal distance between samples (Figure 5). Many widely used DAA methods used in microbiome research support the (Tree)SE framework, including for instance ANCOM-BC2[70], LimROTS[26], MaAsLin3[71], and radEmu[72], and benchdamic benchmarking pipeline[73]. More recently, DPA methods have seen increasing adoption in microbiome research. They ignore abundance data, and focus on comparing feature prevalence i.e. presence/absence profiles. Recent implementations include MaAsLin3 {**(author?)** [71]] and the probabilistic DiPPER[74]. Leveraging the interoperability of this framework, several of these methods have also found applications with other omics modalities, including transcriptomics[75], metabolomics[71], and proteomics[26]. Importantly, recent studies have indicated that elementary methods, such as log-transformed Total Sum Scaled (TSS) counts coupled with a linear model, or presence/absence data with logistic regression, often yield the most consistent results[16;17]. Standard DAA typically focuses on individual features, ignoring interactions or other covariation among feature abundances. Thus, we support covariation analyses. While agglomerating collinear features as a preprocessing step can reduce multiple testing, increase statistical power, and improve interpretability, it is also prone to subjective parameterization and limited generalizability. Features can be grouped based on clustering results from abundance profiles or on external information drawn from phylogenetic relations, taxonomic classifications or functional properties (*e.g.*, aggregating based on metabolite production[51]). Beyond these, multimodal causal mediation analysis for microbiome data have been implemented for SE[76], highlighting the readiness to support the adoption of the rapidly developing multi-omic methodology.

Function- and sequence-level analyses. Taxonomic analysis is frequently complemented by functional analyses, either based on functional predictions derived from amplicon or metagenomic sequences or from parallel metabolomic, metatranscriptomic or other functional assays. Our framework not only supports the incorporation of functional abundance tables within the same data objects, but many statistical methods are generic and can therefore be applied across different omics data types. Where appropriate, specialized methods are also available; for instance, MaAsLin3[71] and many other Bioconductor libraries support metatranscriptomics analysis, notame package[77] for metabolome analyses recently added SE support, and the R for Mass Spectrometry (R4MS) project provides many interoperable tools. The framework has found recent applications in extended analyses of metagenomic features, for instance, antimicrobial resistome composition[65]. We also converted a large-scale compilation of resistome profiles in 8,972 metagenome samples[78] into the TreeSE format to support research and training in this area of microbiome analysis.

Spatio-temporal, ecological, and epidemiological models. Microbiome time series[79] encompass a wide range of study designs, ranging from short to long follow-up times, sparse to dense data, univariate to multivariate and multi-omic monitoring[32;80;81], and real, pseudo, or simulated time series. Moreover, an increasing number of studies incorporate spatial dimension in microbiome analysis[61]. While spatio-temporal analyses require specialized methods, the miaTime package offers a set of data wrangling methods to support longitudinal, multi-modal microbiome analysis. In addition, Bioconductor contains applicable methods from other research domains, such as single-cell omics and spatial transcriptomics, which use closely related data containers. Spatial and temporal information can be incorporated as a covariate (*e.g.*, location or time) or by estimating temporally or spatially resolved metrics, such as divergence (dissimilarity between consecutive time points), autocorrelation (similarity of closely related samples), and feature modality (whether a feature exhibits multiple abundance states). Another type of temporal analysis includes the time-to-event, or survival models, which are encountered regularly in epidemiological and population cohort studies; for microbiome data, these analyses require specialized approaches, as proposed in previous studies[82]. We also provide methods to simulate time series from standard models of microbial community dynamics including Hubbell's neutral model, generalized Lotka-Volterra, and the consumer-resource model through the miaSim package[83]. Our framework provides a robust basis for developing extended methods for longitudinal multi-omics, which is currently an active area of investigation[32]. We also converted a large-scale time series of 16,234 gut microbiome profiles from 585 wild baboons over 14 years[84] into the TreeSE format to support research and training on longitudinal microbiome collections.

Multi-modal integration. A key feature of the ecosystem is to provide the infrastructure supporting the integration of microbial abundance data with other omics modalities, such as transcriptomes, proteomes, or metabolomes[85;86], resistomes[65;78], host genomes[87], or environmental exposomes (Figure 5). By leveraging standardized containers (Fig.~2), users get fluent access to a growing number of multi-omic methods. Containers reduce the need for manual data wrangling and mitigate the risk of errors (*e.g.*, sample mismatches or data formatting), and support modular and reproducible workflows. They also facilitate incorporation of further downstream tools. Multi-omic methods are particularly relevant for the present ecosystem given its enhanced support for integrative analysis. Examples include association-based approaches, latent variable models, and Machine Learning (ML)-based prediction[32;88;89]. Our framework is supported by many common multi-omic libraries in Bioconductor: Cross-correlation analysis, provided by mia, is commonly used to capture direct associations of samples or features between two datasets but it does not explicitly account for potential co-variation between microbial features, and methods such as anansi (Bioconductor package) can additionally incorporate known functional relations into association analyses. Unsupervised latent variable methods such as multi-omics factor analysis (MOFA+)[90] and joint robust PCA (Joint-RPCA)[91] model shared latent structure across modalities, enabling the exploration and visualization of coordinated variation across multiple features and data sources. Supervised methods include the multi-omic

discriminant analysis DIABLO[92] and IntegratedLearner[93], which can leverage dependencies between omics layers towards improved prediction performance. The availability of integrative methods is rapidly increasing, and generic multi-omic toolkits such as mixOmics[94] provide many additional tools. The unified integrative framework that we have adopted can accelerate methods development, benchmarking, and standardization.
Other methods and programming environments. Several other methods and workflows have been employed in microbiome data science. In general, mia supports a flexible statistical programming framework. Thus, it can be used to incorporate more specialized microbiome analysis workflows such as SIAMCAT for association testing[95] or mikropml for machine-learning based prediction and classification[96], network analysis[97], and microbe-set enrichment analysis[98;99]. These tools operate primarily on a single feature matrix, however, and lack the multi-omic feature selection offered by dedicated integration methods discussed in the previous section. The adoption of interoperable containers and shared methodology between related fields can significantly expand the transfer of existing toolkits to adjacent or emerging disciplines. Importantly, our framework can also be linked to external microbiome data analysis pipelines using tools such as bioBakery[47], and foreign platforms through language-specific interfaces, namely Rcpp for C++, reticulate for Python, and JuliaCall for Julia. These connections allow users to combine the most suitable tools across multiple frameworks, while ensuring cross-language interoperability of structured data through standard containers such as (Tree)SE and MAE. The overall data analytical paradigm of Bioconductor is similar to that of scikit-bio[40], a Python-based ecosystem. Whereas scikit-bio benefits from the rapidly evolving machine learning and artificial intelligence tools in Python, Bioconductor's strengths lie in its vast statistical ecosystem and interoperability with dedicated omics research tools. We anticipate that the ongoing developments in cross-language integration will further lower the barriers for cross-utilizing tools across complementary environments.

Accessible analysis and community

Modern science is embodied by a critical community capable of assessing discoveries and replicating results[103]. Open source developer communities present a timely manifestation of this principle, with technical, legal, sociological and epistemic consequences[3]. FAIR principles for software support inclusive access to well-documented and user-friendly methods[1], and are facilitated by the extensive quality control and documentation standards of Bioconductor[27].
In addition to these technical demands, the development and quality control of research software rely significantly on community engagement. This stimulates continuous software improvement and promotes the emergence and dissemination of evidence-based best practices[1;3;79]. Derived quantities such as diversity, prevalence, and other frequently used measures should be calculated early in the workflow, at the data preparation and enrichment step. This helps to avoid unnecessary duplication in the code, mitigating the potential for error and ensuring uniform definitions. Simple approaches should be preferred over complex ones by default, and key analyses should be complemented with visual exploration and verification. These practices can greatly

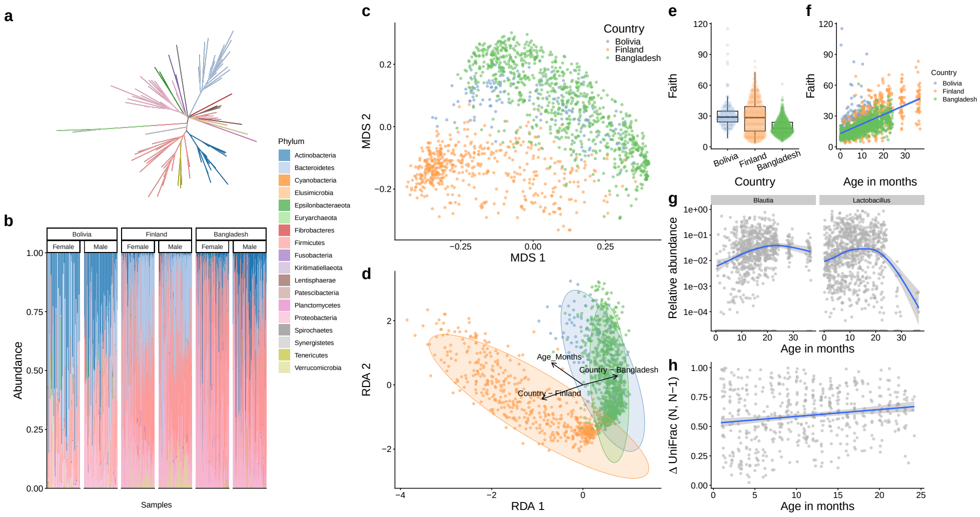

Figure 5: **Common visualization methods for microbiome data.** The selected visualizations demonstrate how the miaViz and scater packages effectively display various aspects of microbiome data, using the SprockettTHData time series dataset openly available via microbiomeDataSets (Bioconductor package). This dataset consists of longitudinal 16S rRNA gut microbiome data from infant fecal samples collected in Finland, Bangladesh, and Bolivia[100–102]. **a**, Phylogenetic relationships. **b**, Phyla abundances of the samples ordered by age, stratified by country and gender. Actinobacteria is more abundant in Bolivia and Bangladesh compared to Finland. **c**–**d**, Multidimensional scaling using UniFrac distances and distance-based Redundancy Analysis (dbRDA) based on Aitchison distance, respectively. Both ordination methods shows clear pattern separating microbial profiles in different countries. dbRDA shows an additional association between microbial profile and age. **e**–**f**, Faith's phylogenetic diversity by country and age respectively. **g**, Differential Abundance Analysis (DAA) results. Two genera exhibit an association with age. **h**, UniFrac divergence between consecutive time points. Results indicate slow stabilization of microbial communities.

improve the transparency and robustness of the analyses and interpretability of the results. Expressive code and meaningful comments support reproducibility and verification.

OMA facilitates standardization across studies by promoting the use of harmonized data structures, validated methods, and informed default settings. This is also facilitating method sharing, benchmarking, and reproducible comparisons. These capabilities help establish community best practices and build consensus while allowing users to flexibly select and evaluate methods appropriate for their specific data and research questions. The Bioconductor framework naturally aligns with these goals through its open ecosystem of data resources, interoperable software, and educational materials. Community building is actively supported through online training resources and communication channels, making it easier for both developers and users to contribute to the continued critical assessment, development and standardization of the ecosystem including methods and documentation. Together with interactive applications, these resources offer accessible entry points for developers and users regardless of their programming experience.

The OMA book. *Orchestrating Microbiome Analysis with Bioconductor* (OMA) is an online book published at https://bioconductor.org/books/release/OMA that provides a comprehensive guide to the (Tree)SE framework for microbiome data science in Bioconductor, including practical guidelines, reproducible workflows, and training material for integrative microbiome analyses. The book is versioned according to the Bioconductor's biannual release cycle. The work incorporates rich feedback from the user community and microbiome training courses. The OMA book aligns to a broader Bioconductor convention, where prominent data structures[27;29;30] designed for multi-table data integration have been described across several domains, such as single-cell omics (OSCA book[24]) and spatial transcriptomics (OSTA Bioconductor book). These resources demonstrate how standard data containers can support statistical programming on interlinked multi-table datasets.

Interactive applications. Graphical user interfaces provide accessible entry points. The iSEEtree and miaDash packages provide interactive analysis and visualization tools for microbiome data[104]. The former extends iSEE to provide support for TreeSE objects, while the latter builds on iSEEtree to enable analyses without requiring any programming. Their interface supports automatic generation of source code and replicable workflows and analysis templates for non-technical users. Moreover, the tidy R paradigm, available through tidyomics, simplifies the syntax and streamlines analytical workflows[105]. These tools allow users to analyze and visualize complex data combinations with ease.

Community. Its widespread adoption has made R one of the most cited software resources of our time[2] and the introduction of the shared data structures facilitates the broad interoperability of the methods ecosystem. Bioconductor has a strong tradition in fostering the R user community through shared data structures and training activities[5]. An active community of developers and users thrives online, supporting the software sustainability and promoting good practices in scientific computing[106]. The community interacts via several online communication channels, including Bioconductor Zulip, GitHub platform, and an email list. The development of the framework is influenced by feedback from the user community through these channels, regular meetings, and training workshops.

Outlook

To enjoy the benefits of automation while also maintaining flexibility, data scientists often recombine open-source software components into custom workflows. However, the lack of commonly adopted data representation standards can hinder the development, sharing, and application of modular, interoperable statistical workflows. Here, we present an R/Bioconductor ecosystem for multi-modal microbiome data science. Building upon previously established data structures[29;30], we have introduced a comprehensive, interoperable methods ecosystem extending these, developed as a coordinated community effort. We also provide exhaustive guidance in the OMA book, thereby supporting independent learning and reproducible analysis in microbiome research and beyond.

The Bioconductor community is widely recognized for developing open research software for life sciences. Its strengths include minimal documentation standards, portability across operating platforms, quality control (*e.g.* peer-review, unit testing, and continuous integration), usability (APIs), robustness (dependency testing and deprecation mechanisms), and a biannual release cycle with semantic versioning. Over the past two decades Bioconductor has grown into a repository of more than 2,300 actively maintained software packages, of which 62 are now classified under the *Microbiome* Task View. At the core of the Bioconductor's data science strategy has been the idea of shared data representations[4;31]. This has enhanced interoperability, mitigated redundant efforts and facilitated the translation of methods between research domains[1;107]. In addition to these technical advances, the shared standards have played an important role in the formation of broad developer and user communities[5].

Microbiome research has shifted towards integrative analyses linking various omics towards enhanced mechanistic and causal insights[13;108]. Technological advances in long-read sequencing, functional assays, spatial profiling, and single-cell analysis have created an increasing demand for integrative analysis techniques that can link microbiome data with other omics along varying spatial and temporal dimensions. In concert with parallel developments in other omics, our framework specifically addresses the need to integrate and jointly analyze interlinked abundance tables and supporting hierarchical and tabular side information across different omics. Whereas our framework supports the basic data wrangling and microbiome data science operations, it also provides the structured input necessary for probabilistic inference and scalable computation. This is essential as the recent developments in microbiome and multi-omics workflows increasingly rely on advanced statistical and ML techniques such as Bayesian modeling and Gaussian processes to accommodate uncertainty and temporal or spatial structures. Building upon these statistical foundations, the rapid emergence of foundation models and deep learning approaches in biomedicine calls for robust, scalable data representations such as those we have adopted. Applications include transfer learning across studies, interpretable embeddings of microbiome multi-omic profiles, computational prediction of microbial function and metabolites[109;110], and integration with AI agents for hypothesis generation and automated analysis, among others.

Open data science infrastructures have become critical components of reproducible research and microbiome applications critically rely on open-source methods and workflows[1;3;79]. Standardized data containers can enhance the transparency, trustworthiness, and explainability of analyses. We are documenting the open data science ecosystem for multi-modal microbiome research built on widely adopted Bioconductor data containers, software libraries, data resources, tutorials, and a growing community of developers. The field can benefit from improved support for the FAIR data principles[111]; this could be facilitated by adding built-in refinement, enrichment, and validation methods for the relevant community standards[112–114]. The users could benefit from a collection of end-to-end case studies demonstrating how different example analyses can be conducted in practice, from original source data to results; we welcome contributions of well-documented workflows demonstrating the use of the mia framework. Finally, supporting interoperability between R and other programming languages could facilitate the application of other ML frameworks. As the field evolves, the developer communities will continue to integrate new tools and approaches to improve integrative analyses of microbiome data.

Data availability

All datasets used in this article are publicly available with a permanent digital object identifier and an open source license from the specified sources.

Code availability

This article presented v1.1.1 of the OMA book (2026-05-07) published at https://bioconductor.org/books/release/OMA, which depends on R v4.6 and Bioconductor v3.24. Analysis scripts and results of the benchmarking analysis are available on Zenodo (mia community).

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Author contributions

L.L. and T.Bo. conceived and coordinated the study. L.L., T.Bo., G.B., A.R., A.S., R.H. and T.Ba. performed the data analysis. L.L., T.Bo. and G.B. wrote the original draft. L.L. acquired funding. All authors contributed material, participated in manuscript preparation and approved the final version.

Competing interests

The authors declare no competing interests.

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